Mulching: soil cover with organic material

Spatial scale: vineyard

About this sustainable management strategy

Mulching is a vineyard management practice in which the soil surface beneath or between vines is covered with a layer of organic material. Common mulching materials include straw, hay, compost, leaves, bark, and shredded vineyard residues such as pruning cuttings and wood chips. By covering the soil surface, mulches help suppress weed growth, reduce water evaporation, moderate soil temperatures, and protect the soil from erosion.
As organic mulches gradually decompose, they contribute organic matter and nutrients to the soil, supporting soil fertility and biological activity. Mulching can also improve soil structure, increase water-holding capacity, and enhance vineyard resilience to drought and extreme weather conditions. The choice of mulch material, application rate, and timing should be adapted to local soil, climate, and vineyard management objectives.

Key ecosystem services

- Soil health and fertility (*improved soil structure, nutrient cycling and organic matter content*)
- Erosion control *(slope stabilisation)*
- Water retention *(reduced surface runoff)*
- Water regulation *(improved infiltration and soil moisture conservation)*
- Microclimate regulation (*Soil temperature control*)
- Biodiversity enhancement (*above-and below-ground*)
- Weed control
- Grape production *(improved vine growth and resilience under water-limited conditions)*

Considerations for successful implementation

Pest, disease and phytosanitary management

The selection and management of mulching materials should consider local pest and disease pressures. Regular monitoring and the use of suitable organic materials can help maximise soil benefits while minimising the potential for pests or pathogens to establish within the mulch layer.

Management and labour requirements

The application and maintenance of mulch require planning and regular management, particularly regarding material sourcing, application timing, and replenishment. Integrating mulching practices into existing vineyard operations can support their long-term effectiveness and facilitate adoption.

Technical requirements and site-specific performance

The effectiveness of mulching depends on factors such as soil type, climate, mulch material, and application rate. Regular soil and vine monitoring, combined with site-specific adjustments, can help optimise soil moisture conservation, nutrient cycling, and erosion protection while maintaining healthy vine growth.